

reference format

## Font format: Arial, Front size:11

Date:

Title:

Purpose:

Procedure:

1. Heat the top of the glass vial with the Bunsen burner.
2. Add sterile water and crack it with tweezers.
3. Remove paper and cotton.
4. Add 500  $\mu\text{L}$  LB medium and resuspend the tablet.
5. Drop the supernatant and streak it with four-quadrant streak method.
6. Spread 100  $\mu\text{L}$  supernatant on LB plate with 100  $\mu\text{L}$  25 mg/mL chloramphenicol.
7. Inoculate the remaining supernatant in 5 mL LB medium with 25  $\mu\text{g/mL}$  chloramphenicol.
8. Incubate in 37°C, started from 18:00

Done by Chars, recorded by Chars.

Results:

## Common mistakes.

1. Chloramphenicol
2. pNW33N-L2 (sfGFP)
3. Autoclave for 20 minutes at 127°C.
4. *Bacillus subtilis*

## Make format consistent.

1. degrees Celsius->°C
2. ul->  $\mu\text{L}$
3. 23444xg->23444 g
4. ddH<sub>2</sub>O->ddH<sub>2</sub>O
5. OD600



2026/5/1



**Date: 20260501**

**Title: Medium Preparation**

**Purpose: Prepare medium for *Bacillus Subtilis* electroporation**

**Procedure:**

A. Growth Medium (GM medium) :

1. Prepare medium in 500mL flask

| substance                                 | amount        |
|-------------------------------------------|---------------|
| LB broth powder                           | 6.25 g        |
| D - sorbitol (final concentration: 0.5 M) | 22.77g        |
| Deionized water                           | Add to 250 mL |

2. Stir on the heat plate until dissolved
3. Autoclave for 20 min

B. Electroporation buffer (EP buffer)

1. Prepare medium in 500 mL flask

| substance       | amount        | Final concentration |
|-----------------|---------------|---------------------|
| D - sorbitol    | 22.77 g       | 0.5 M               |
| D - mannitol    | 22.77 g       | 0.5 M               |
| Glycerol        | 25 mL         | 10%                 |
| Deionized water | Add to 250 mL |                     |

2. Stir on the heat plate until dissolved
3. Autoclave for 20 min

Done by Chars, recorded by Alex

2026/5/2



**Date: 20260502**

**Title: Plasmid Transformation**

**Purpose: Pick up single colony of pNW33N-L2 (sfGFP) C1 and G1 for sequencing**

**Procedure :**

1. Prepare the following substances in two separate microtubes:

|                       | microtube#1                                                                        | microtube#2                                                                         |
|-----------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| DH5α                  | 30 µL                                                                              | 30 µL                                                                               |
| pNW33N-L2 (sfGFP) #C1 | 2 µL                                                                               |  |
| pNW33N-L2 (sfGFP) #G1 |  | 2 µL                                                                                |

2. Incubate the competent cells on ice for 20 minutes.
3. Heat shock at 42°C for 45 seconds
4. Put them back on ice immediately for 3 minutes
5. Transfer 30 µL of the transformants onto each of two LB agar plates containing chloramphenicol.
6. Spread evenly on agar plate
7. Place the two agar plates in the incubator for 37°C overnight

Done by Sophie H. recorded by Sophie H.

2026/5/3



**Date: 20260503**

**Title: Pick up single colony of sfGFP #G1**

**Purpose: To amplify the pNW33N-L2 (sfGFP) G1 clone from a bacterial colony on an LB agar plate**

**Procedure:**

1. Place 5 ml LB broth into each 15ml of falcon tube (#1, #2, #3, and #4)
2. Add 20  $\mu$ L chloramphenicol in each falcon tubes
3. Pick four individual colonies of the pNW33N-L2 (sfGFP) G1 clone and inoculate each colony into a separate tube (#1–#4).
4. Place the four falcon tubes in the incubator and incubate at 37°C overnight with shaking at 200 rpm

Done by Sophie H, Recorded by Sophie H.

2026/5/4 

**Date: 20260504**

**Title: Miniprep & Glycerol stock**

**Purpose: Isolate the pNW33N-sfGFP #G plasmid and store the glycerol stock for long-term storage of plasmids**

**Procedure:**

1. Collect 4 mL of cells by centrifuge at 7000 g for 1 minute in 1.5 mL tubes
2. Mix 0.5 mL culture medium with 50% glycerol into the screw-top tubes:
  - a. Tube 1 x2
  - b. Tube 2 x 2
  - c. Tube 3 x 2
  - d. Tube 4 x 2
3. The tubes were then freezed at -80 degree
4. Add 200 µL MX1 buffer and resuspend the cells by pipetting
5. Add 250 µL MX2 buffer and mix the solution by gently inverted the tubes 8 times
6. Add 350 µL MX3 buffer and mix it by inverted 8 times
7. Centrifuge the tubes at 23444 g for 5 minutes
8. Transfer the supernatant into the column, centrifuge at 7000 g, 1 min and discard the flow through
9. Add 500 µL WN buffer and centrifuge at 7000 g for 1 min and discard the flow though
10. Discard the flow through
11. Add 700 µL WS buffer and centrifuge at 7000 g for 1 min
12. Discard the flow through
13. Centrifuge the column at 23444 g for 5 min
14. Add 50 µL EB into the center of membrane
15. Place at RT for 20 min
16. Centigrade at 23444 g for 2 min
17. Measure the concentration

Done by Gabriel

**Date: 20260504**

**Title: electroporation competent B.sub 168 preparation**

**Purpose: To make the electrocompetent cells for electroporation transformation**

**Procedure:**

1. Dilute an overnight culture of B.sub 168 16-fold in growth medium and grow at 37°C to 0.600 of 0.7 (5mL +313 µL B.sub)  
Growth medium : LB medium containing 0.5M sorbitol
2. Cool the cells on ice for 10 min with shake
3. Harvest the cells by centrifugation at 4 °C and 5000 g for 5 min
4. Wash the cells by adding 1 mL ice-cold electroporation medium two times  
Electroporation medium : 0.5 M sorbitol , 0.5 M mannitol and 10% glycerol
5. Suspend the cell in 1/40 of the culture volume (5mL) of electroporation medium  
(5mLX1/40=125µL EP buffer)
6. Stored the cells at -80°C

Done by Gabriel

2026/5/6



**Date:20260506**

**Title: Prepare Salinity Agar Plate**

**Purpose: Prepare salinity plate for testing trehalose rescue experiment**

**Procedure:**

1. Prepare mixture through following the recipe below:

|                     | Control  | 50 mM NaCl | 50 mM NaCl<br>with 1 mM<br>trehalose | 100 mM NaCl | 100 mM NaCl<br>with 1 mM<br>trehalose |
|---------------------|----------|------------|--------------------------------------|-------------|---------------------------------------|
| ½ MS<br>Medium      | 2.37 g   | 1.975 g    | 1.975 g                              | 1.975 g     | 1.975 g                               |
| NaCl                | –        | 0.1461 g   | 0.1461 g                             | 0.2922 g    | 0.2922 g                              |
| 100 mM<br>trehalose | –        | –          | 500 µL                               | –           | 500 µL                                |
| Deionized<br>water  | To 60 mL | To 50 mL   | To 50 mL                             | to 50 mL    | to 50 mL                              |

2. Autoclave in 127°C for 20 minutes
  3. Dispense in 10 cm petri dish and wait until it solidify
  4. Sow 9 arabidopsis seeds in each plate
  5. Vernalization at 4°C, and wrapped with aluminium foil to avoid from light
- Done by Chloe, supervised by Chars

2026/5/7



**Date: 20260507**

**Title: Miniprep**

**Purpose: Prepare plasmid pWN33N-L2(sfGFP) for *Bacillus Subtilis* electroporation**

**Procedure:**

1. Collected the cells from 20 mL TB medium liquid culture by centrifuge at 23444 g for 1 min until collected the liquid culture in 4 x 2 mL micro tube.
2. Resuspend in 250  $\mu$ L Buffer P1. Note: We used Qiugen miniprep instead of Viogen miniprep.
3. Add 250  $\mu$ L Buffer P2 and invert the tube 10 times.
4. Add 350  $\mu$ L Buffer N3 and invert the tube 10 times.
5. Centrifuge at 23444 g for 10 min.
6. Transfer 800  $\mu$ L supernatant to column. Repeat this step until all of the supernatant is collected.
7. Centrifuge at 17900 g for 1 min.
8. Remove the flow-through. Add 500  $\mu$ L Buffer PB.
9. Centrifuge at 17900 g for 1 min. Remove the flow-through.
10. Add 750  $\mu$ L Buffer PE.
11. Centrifuge at 17900 g for 1 min. Remove the flow-through.
12. Centrifuge the empty column for 1 min.
13. Transfer the column to a new 1.5 mL micro tube.
14. Add 50  $\mu$ L nuclease-free water ( $\sim 65^{\circ}\text{C}$ ). Incubate at RT for 5 min.
15. Centrifuge at 17900 g for 3 min.
16. Measure the concentration by Biodrop.

|                     | Concentration (ng/ $\mu$ L) | A260/230 | A260/280 |
|---------------------|-----------------------------|----------|----------|
| pNW33N-L2(sfGFP) #A | 56                          | 2.24     | 1.867    |
| pNW33N-L2(sfGFP) #B | 71                          | 2.44     | 1.919    |

Done by Chars, Recorded by Sophie L

**Date:20260507**

**Title:Prepare recovery medium**

**Purpose: Prepare recovery medium for *Bacillus Subtilis* electroporation**

**Procedure:**

1. Prepare the medium by following the recipe below:

|                  | Weight (g) | Final concentration |
|------------------|------------|---------------------|
| SOC broth powder | 1.585      | 1X                  |
| D-sorbitol       | 4.55       | 0.5 M               |

|                 |                 |        |
|-----------------|-----------------|--------|
| D-mannitol      | 3.46            | 0.38 M |
| Deionized water | Top up to 50 mL |        |

2. Filter through a 0.22  $\mu\text{m}$  filter

Done by Chloe & Sophie H, supervised by Chars

2026/5/9



**Date: 20260509**

**Title: Prepare ½ MS agar (30 mL)**

**Purpose: Prepare agar for hydroponics cultivation of *Arabidopsis* Procedure:**

1. Prepare the mixture through the following recipe:

|                    | Weight (g) |
|--------------------|------------|
| ½ MS Medium        | 0.069      |
| ddH <sub>2</sub> O | 30 ml      |

2. Add stir bar into mixture and place the beaker with the mixture on the heat plate.
3. Cover the beaker with aluminum foil and tape a piece of autoclave tape.
4. Autoclave for 20 min at 127°C.
5. Turn on the Biological Safety Cabinet.
6. Lift sash to operating height.
7. Wipe the working space with 70% ethanol.
8. Spray Petri dishes, micropipette, and beaker (contain lids) with 70% ethanol and put into the cabinet.
9. Exposure of UV for 10 mins, with sash closed.
10. Raise the sash to the certified operating height.
11. Spray hands with 70% ethanol and place lids in each dish.
12. When the autoclave ends, waterbath the agar medium to cool down.
13. Micropipette 200 µL into each cap.
  - a. Ensure all caps are placed flat on the dish.
14. Finish 3 plates of caps.

Done by Abby T, Supervised by Gina, Recorded by Sophie L

2026/5/10 

**Date: 20260510**

**Title: Electroporation**

**Purpose: Testing the condition of electroporation for *Bacillus Subtilis***

**Procedure:**

1. Prewarm SOC- recovery Medium (SOC-RM) in 37°C and pre-chill cuvette
2. Mix pNW33N-L2(sfGFP) (54 µg/ml) 190 µl with electrocompetent cell 90 µl
3. Transfer the plasmid-competent cell mixture into pre-chill 0.2 cm cuvette Setup electroporator

a.

|         | Voltage (kV) | Capacity (µF) | Resistance (Ω) | TC (ms) | Notes      |
|---------|--------------|---------------|----------------|---------|------------|
| Trial 1 | 0/2.0        | 25            | ∞              | 0       | Arc        |
| Trial 2 | 0/1.8        | 25            | ∞              | 0       | Arc        |
| Trial 3 | 1.995/2.0    | 25            | ∞              | 6553.5  | Dilute 1/2 |
| Trial 4 | 2.195/2.2    | 25            | 300            | 6.7     |            |
| Trial 5 | 2.091/2.1    | 25            | 300            | 6.7     |            |

4. Add 1 ml of SOC-RM into cuvette and transfer to 15 mL tube
5. Incubate at 37°C and 200 rpm for 4 hours
6. Trial 3,4,5 were used for the subsequent experiment
7. Prepare three LB agar plates with chloramphenicol and three without chloramphenicol.
8. Prepared the diluted samples 1:1,000,000 for each one
9. The remaining samples (original) were centrifuged at 5000 g for 5 minutes
10. The dilute samples were spread on LB plate
11. The original samples were spread on LB plate with chloramphenicol
12. Incubate the plate at 37°C overnight

Done by Gabriel

2026/5/11



**Date: 20260511**

**Title: Preparation of *Bacillus subtilis* competent cells**

**Purpose: Prepare electrocompetent *Bacillus subtilis* cells that can efficiently take up foreign DNA during electroporation.**

**Procedure:**

1. Dilute an overnight culture of *Bacillus Subtilis* in 16 fold in growth medium (LB medium containing 0.5 M sorbitol) (312.5  $\mu$ L of *Bacillus Subtilis*) in 5 ml Growth medium and grow at 37°C to achieve OD600 to 0.944
2. Cool the cells on ice cold water and put on shaker for 10 minutes
3. Centrifuge at 4°C at 5000 g for 5 mins to harvest cells (total of 4.5 mL)
4. Add 1 mL ice cold electroporation medium to resuspend the cells in 1.5 mL microtube, centrifuge 5000 g for 1 minute and discard supernatant
5. Repeat step 4 for a total of 4 times to wash cells
6. Suspend the cells in 1/40 of the culture volume of the electroporation solution (0.1125 mL) with a cell concentration of  $1-1.3 \times 10^{10}$  cfu/mL
7. Store competent cells at -80°C

**Problems encountered:** Step 4 produced small pellets so centrifuged for 5 mins instead of 1 minute

Done by Chloe, documented by Olivia Lin

2026/5/12

**Date: 20260512**

**Title: Purification of plasmids**

**Purpose: Check if *Bacillus subtilis* electroporation (transformation) was successful**

**Procedure:**

**I. Plasmid preparation(pNW33N-L2 (sfGFP))**

1. Pick up single colonies and transfer each to 5 ml of LB medium
2. Add 5  $\mu$ L of chloramphenicol (25 mg/ml) to each
3. Let the culture incubate for 25°C overnight

**II. Purification of plasmids (applies to all 7 plasmids)**

1. Transfer 2 ml of the overnight culture to a 2ml microtube
2. Centrifuge at 16000 g for 1 min
3. Discard the supernatant
4. Make the MX1 buffer mix consisted of MX1 buffer and lysozyme

**4-1 Calculations for 7 tubes:**

-Amount of MX1:  $250 \mu\text{L} \times 9 = 2250 \mu\text{L}$

-For 1mg/ml concentration of lysozyme:  $2250 \text{ mg} / 1000 = 2.5 \text{ mg}$

-For 10mg/ml concentration of lysozyme:  $2.25 \text{ mg} / 10 = 0.225 \text{ ml} = 225 \mu\text{L}$

-Buffer:  $2250 \mu\text{L} - 225 \mu\text{L} = 2025 \mu\text{L}$

5. Add 200  $\mu$ L of the MX1 buffer mix
6. Incubate at 37°C for 10 mins
7. Resuspend the pellet by pipetting
8. Add 250  $\mu$ L of MX2 buffer
9. Gently mix the solution by inverting the microtube 8-10 time
10. Incubate the mixture at RT for 2 min
11. Add 350  $\mu$ L of MX3 buffer
12. Gently mix the solution immediately
13. Isolate the supernatant with plasmid and centrifuge it at 16000 g for 5 min
14. Transfer the supernatant to column and centrifuge at 7000 g for 30 sec (discard the flow through)
15. Add 500  $\mu$ L of WN buffer, centrifuge at 7000 g for 30sec and discard the flow through
16. Add 700  $\mu$ L of WS buffer, centrifuge at 7000 g for 30sec and discard the flow through
17. Centrifuge the empty column at 10000g for 3min to remove residual ethanol
18. Place the column into a new 1.5mL microtube
19. Add 50  $\mu$ L of elution buffer on the center of the membrane
20. Incubate the column at RT for 3 min then centrifuge at 10000 g for 3 min
21. Measure the concentrations with Biodrop

**Result:**

| Sample | Concentration (mg/mL) | A260/280 | A260/230 |
|--------|-----------------------|----------|----------|
| #1     | 21                    | 1.909    | 0.913    |
| #2     | 18                    | 1.800    | 1.125    |
| #3     | 11                    | 2.200    | 5.500    |
| #4     | 16                    | 2.000    | 2.667    |
| #5     | 27                    | 1.929    | 1.125    |
| #6     | 16                    | 2.000    | 5.333    |
| #7     | 20                    | 1.667    | 0.500    |

The DNA concentration will thus be used to calculate the reaction formula of enzyme digestion

Done by Sophie C. Recorded by Sophie H.

**Date:** 2026/5/12

**Title:** pNW33N-L2 (sfGFP) plasmid linearization

**Purpose:** To prepare the purified pNW33N-L2 (sfGFP) plasmids for gel electrophoresis analysis

**Procedure:**

1. Calculate reaction formula:

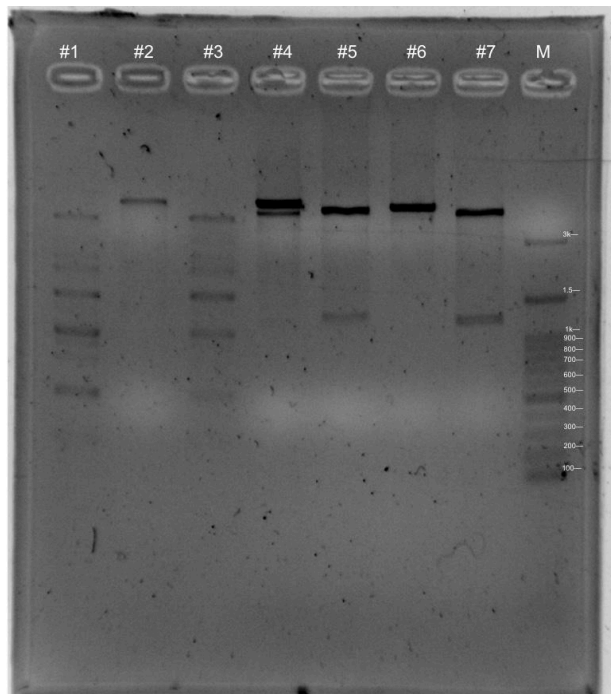
| Sample | DNA  | ddH <sub>2</sub> O | EcoRI | DNA1 $\mu$ L | ddH <sub>2</sub> O | BtsI |
|--------|------|--------------------|-------|--------------|--------------------|------|
| #1     | 11.9 | 5.6                | 0.5   | 11.9         | 2.8                | 1    |
| #2     | 13.9 | 3.6                | 0.5   | 13.9         | 1.8                | 1    |
| #3     | 20   | 0                  | 0.5   | 20           | 0                  | 1    |
| #4     | 15.6 | 1.9                | 0.5   | 15.6         | 0.95               | 1    |
| #5     | 9.3  | 8.2                | 0.5   | 9.3          | 4.1                | 1    |
| #6     | 15.6 | 1.9                | 0.5   | 15.6         | 0.95               | 1    |

|                                                |      |   |     |      |     |   |
|------------------------------------------------|------|---|-----|------|-----|---|
| #7                                             | 12.5 | 5 | 0.5 | 12.5 | 2.5 | 1 |
| +10X rCutSmart buffer in each tube : 2 $\mu$ L |      |   |     |      |     |   |

\*Samples 1~7 are plasmids from 5/12's purification under the conditions of 2.2 kv, 300 $\Omega$ .  
And the plasmid is pNW33N-L2 (sfGFP)

2. Prepare reaction by adding ddH<sub>2</sub>O and buffer first, then add DNA and enzymes
3. Centrifuge the 7 microtubes by balancing 6 microtubes for 1st round of centrifuge and keep one microtube with the one left to balance for second round to centrifuge(each round 5 sec)
4. Put microtubes in PCR machine at 37°C for an hour to incubate
5. Electrophoresis
  - a. Add and measure 0.3 grams of agarose powder
  - b. Measure 30 ml of TAE buffer in measuring cylinder
  - c. Wrap flask with plastic (food) wrap on the opening
  - d. Use pipette tip to poke 4~5 holes in the wrap
  - e. Place flask in microwave for 30 sec (wear cotton gloves)
  - f. Take out the flask and gently shake to see if the mixture is clear and all the powder has dissolved.
  - g. If not, microwave for another 30 sec
  - h. Assemble the tray (8 slots: 1 DNA ladder, 7 samples)
  - i. Pour agarose mixture in tray
  - j. Use the large tip to press the bubbles in the mixture/tray away
  - k. Wait for gel to solidify and take gel into electrophoresis tray containing the TAE buffer
  - l. Take samples out of the PCR machine
  - m. Aliquot 2  $\mu$ L of 6 X loading dye onto the parafilm for each sample. Pipette
  - n. Pipette and mix 10  $\mu$ L of each sample with the dye. Then, load 10  $\mu$ L of the mixture into each well.
  - o. Connect the electrodes (black-black, red-red) and run the gel at 100 V , check for bubbles in the buffer and make sure the samples are moving toward the positive electrode.

## Results:



### 5/12 Plasmid Linearization by enzyme digestion

M: 1kb DNA ladder  
#1-#5: pNW33N-L2(sfGFP)

Sample #1-#5 are from 5/12's purification under the conditions of 2.2 kv, 300Ω.

Gel electrophoresis of Linearized pNW33N-L2(sfGFP) plasmid  
Done by Abigail, Abby Tsai, Recorded by Sophie H.

2026/5/13

**Date: 20260513**

**Title: Colony PCR**

**Purpose: Verify the results of the electroporation**

**Procedure:**

1. Prepare the master mix

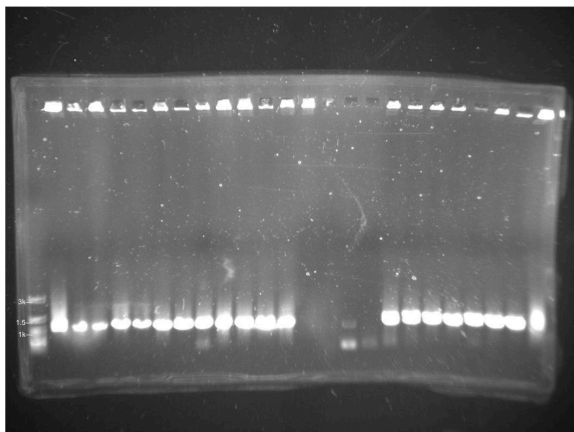
|                    |            |             |                                |
|--------------------|------------|-------------|--------------------------------|
|                    | 1 rxn      | 25 rxn      |                                |
| Forward            | 1 $\mu$ L  | 25 $\mu$ L  | FW primer :<br>BioBrick prefix |
| Reverse Primer     | 1 $\mu$ L  | 25 $\mu$ L  | RV primer:<br>BioBrick suffix  |
| Q-Amp Master Mix   | 10 $\mu$ L | 250 $\mu$ L |                                |
| ddH <sub>2</sub> O | 8 $\mu$ L  | 200 $\mu$ L |                                |

2. Dispense reaction to 8-strip PCR tube
3. Pick up the colonies from the 2.0 kv, 2.1 kV, 2.2 kV condition plate
4. React in the following thermal cycle

|      |       |           |
|------|-------|-----------|
| 95°C | 5min  | 1 cycle   |
| 95°C | 30sec | 35 cycles |
| 55°C | 30sec |           |
| 72°C | 1min  |           |
| 72°C | 5min  | 1 cycle   |

5. Prepare a 1% agarose gel
6. Resolved PCR product in 1% agarose gel

**Result:**



Done by Chloe, Olivia Lin, Abigail

Documented by Olivia Lin

2026/5/14 

**Date: 20260514**

**Title: Miniprep**

**Purpose: Prepare the shuttle vector pNW33N-L2 (sfGFP) for electroporation**

**Procedure:**

1. Transfer 1500  $\mu$ L of overnight liquid culture inoculated from colonies on the streak plates used for the colony (Done in 5/13) to each tube (total 8 tubes)
2. Centrifuge for 3 min at 6800 g, 22°C
3. Aspirate supernatant from 8 tubes with micropipette
4. Add the rest of the liquid culture
5. Centrifuge at 22°C 23944 g for 1 minute
6. Collect supernatant transfer to column
7. Centrifuge at 17900 g for 30 seconds
8. Repeat this step, until collect all the supernatant
9. Add 500  $\mu$ L PB buffer and centrifuge at 17900 for 30 seconds.
10. Discard the flow-through at 750  $\mu$ L PE buffer
11. Centrifuge at 17900 g for 3 minutes
12. Discard the flow through and centrifuge at 17900 g for 3 minutes.
13. Transfer the column to new 1.5 ml microtube
14. Add 50  $\mu$ L nuclease free water at 65°C
15. Incubate at RT for 3 minutes
16. Centrifuge at 17900 g for 3 minutes
17. Measure the concentration

|                      | Concentration (ng/ $\mu$ L) | A260/230 | A220/280 |
|----------------------|-----------------------------|----------|----------|
| pNW33N-L2 (sfGFP) #1 | 53.0                        | 1.606    | 1.893    |
| pNW33N-L2 (sfGFP) #2 | 142.0                       | 2.152    | 1.919    |

Done by Chloe, Olivia Lin, Ryan Yuan

**Date: 20260514**

**Title: Preparing NaCl or Trehalose containing agar plate -> remake 20260506's salinity plate**

**Purpose: Prepare the mixture for new salinity plates**

**Procedure:**

1. Prepare the mixture by following the recipe below

|  |         |            |                             |            |                               |
|--|---------|------------|-----------------------------|------------|-------------------------------|
|  | control | 500mM NaCl | 50mM NaCl / 0.5mM trehalose | 100mM NaCl | 100mM NaCl w/ 0.5mM trehalose |
|--|---------|------------|-----------------------------|------------|-------------------------------|

|                  |         |         |         |          |          |
|------------------|---------|---------|---------|----------|----------|
| ½ MS medium      | 1.975 g | 1.975 g | 1.975 g | 1.975 g  | 1.975 g  |
| NaCl             |         | 0.1461g | 0.1461g | 0.2422 g | 0.2922 g |
| 100mM trehalose, |         |         | 25 µL   |          | 250 µL   |
| deionized water  | 50 ml   | 50 ml   | 50 ml   | 50 ml    | 50 ml    |

2. Autoclave at 127°C for 20 mins
3. Dispense in 9 cm Petri dish, dry for 1 hour
4. Place 9 Arabidopsis seeds in each plate in a grid pattern
5. Vernalization at 4°C, avoid light
6. Add 30 ml of deionized water and stir with hot plate before adding the rest to 50 ml
7. Autoclave at 127°C for 20 mins
8. Arrange the concentration to 100 ml and 150 ml
9. Add a control with trehalose and no sodium chloride

Done by Abby Kao, Abigail, Audrey H, Naomi

2026/5/16 

**Date: 20260506**

**Title: Preparing Hydroponics Medium**

**Purpose: Prepare MS medium with adjusted pH levels for hydroponic system**

**Procedure:**

1. Prepare the mixture by following the recipe below.
  - a.  $\frac{1}{2}$  MS medium : 4.6 g / 2L
  - b. ddH<sub>2</sub>O : 2 L
2. Pour about 900 mL of ddH<sub>2</sub>O inside the beaker and stir it with the MS medium powder using a magnetic stir bar.
3. Before using the pH meter, use pH 4 and pH 7 buffer solutions to do two-point calibration.
4. Use the pH meter to determine the pH of the medium. Adjust pH by adding KCl. Record the before and after pH results.
5. Fill the rest of the ddH<sub>2</sub>O into the beaker.
6. Distribute 2 L of the mixture to five 500 mL flasks, each flask is filled with 400 mL of medium mixture.
7. Cover up the opening of all flasks with foil and stick the autoclave indicator tape on the foil.
8. Pour 1100 mL of water to the autoclave and autoclave the five flasks for 20 minutes.
9. Use tap water to cool down each flask.
10. After cooling, fill the hydroponics box with medium.
11. Transfer the plant seed to the hydroponic box.

**Result:**

pH results of mixture:

- Before pH result: pH 4.05
- After pH result: pH 5.62

Done by Naomi Lin, Joshua Hong, Sara Chen, Sophie Huang

Recorded by Naomi Lin, Sara Chen

2026/5/17 ✓

**Date:20260517**

**Title: Electrophoration**

**Purpose: Testing the condition for electroporation - remake 5/10 to increase colony number**

**Procedure:**

1. Prewarm SOC-recovery medium (SOC-RM) in 37°C and precold cuvette.
2. Mix pNW33N-L2 (sfGFP) (142 ng/mL) 7 µL and electrocompetent cell 93 µL
3. Pre-chill 0.2 cm cuvette with plasmid-competent cell mixture
4. Set up electroporator

|         | Voltage (kV) | Capacity (µF) | Resistance (Ω) | TC (ms) |
|---------|--------------|---------------|----------------|---------|
| Trial 1 | 1.977/2.0    | 25            | 200            | 5.05    |
| Trial 2 | 2.074/2.1    | 25            | 200            |         |
| Trial 3 | 2.182/2.2    | 25            | 200            | 4.6     |

5. Add 1 ml of SOC-RM to each tube, transfer to 15 mL tubes. Don't fasten the tube completely, tape it to fix the lid of the tube.
  6. Incubate at 37°C, 200 rpm for 4 hrs
  7. Put the LB plate in the incubator
  8. Dilute plasmid with LB medium with the ratio of 1:100,000
    - a. First dilute 1:1000 (1 µL:999 µL), then 1:100 (1 µL:99 µL)
  9. Use the applicator stick to spread the diluted plasmid solution over LB plate
  10. Take the rest of the 1:100 mixture and centrifuge it at 5000 g for 5 minutes.
  11. Discard the supernatant and leave the pellet.
  12. Resuspend the tubes.
  13. Spread 100 µL of the LB culture onto a chloramphenicol-containing LB agar plate.
  14. Incubate overnight
- Done by Sara Chen, Sophie Huang, Chloe Wu, Abigail

**Date: 20260517**

**Title: Prepare 1 Liter of LB Agar Plate**

**Purpose:Prepare LB agar plate**

**Procedure:**

1. Clean the flasks with tap water followed by followed by rinsing ddH<sub>2</sub>O, add magnetic stir bar
2. Measure 25 g of LB, 15 g of Agar (3.75 g / 250 mL), and add 1000 ml of ddH<sub>2</sub>O into the flask
3. Distribute 500 mL to two separate 500 mL flasks.
4. Cover each flask opening with foil and stick autoclave indicator tape on top of the foil. Autoclave for 20 minutes.

Done by Chloe W, Ryan Wei, Ryan Yuan, Abigail, Naomi, Phoebe Chen, Alex

Recorded by Naomi



2026/5/18



**Date: 20260518**

**Title: Remaking *Bacillus subtilis* Competent Cell**

**Purpose: To prepare for electroporation**

**Procedure:**

1. Dilute the original bacterial culture at a 1:16 ratio and distribute it into six 10-mL tubes, for a total volume of 60 mL.
2. Grow the bacteria culture at 37°C to an OD600 of 0.944
3. Take 2 ml of the culture for testing absorbance
4. Repeat step 1 & 2 until the culture's absorbance OD600 is 0.85-0.95
  - a. Trial 1: OD = 0.67 (after 2 hours)
  - b. Trial 2: OD = 1.205 (after 1 hour; could adjust time to 30 mins)
5. Cool the diluted tubes for 10 minutes in ice bath
6. Place the ice bath on the shaker to cool the solution quickly
7. Centrifuge 5000 g at 4°C for 5 minutes
8. Prepare and continue lab work under aseptic operating conditions
9. Add a total of 4.8 ml cells into a microtube
10. Add 1ml of cold Electroporation Buffer
11. Pour out the supernatant
12. Add 1 ml of cold Electroporation Buffer
13. Remove the supernatant
14. Resuspend the cells with a 40:1 ratio of 4.8 ml (add 1.2 ml or 0.12 ml)
15. Place the microtube in the -80°C fridge overnight

Done by: Chloe , Sara C, Sophie H, Abigail L, Renee K, and Ethan C

2026/5/19



**Date: 20260519**

**Title: Electroporation**

**Purpose: Testing the conditions by which has the best transformation efficiency on *Bacillus subtilis***

**Procedure:**

1. Rewarm the recover medium at 37°C
2. Mix pNW33N-L2 (sfGFP) (142 ng/ml) 7 microliters to 93 microliters of competent cells (5/18)
3. Pre-chill 0.2cm cuvette with plasmid competent cell mixture
4. Set up electroporator

|         | Voltage (kv) | Capacity ( $\mu$ F) | Resistance ( $\Omega$ ) | TC (ms) |
|---------|--------------|---------------------|-------------------------|---------|
| Trial 1 | 2.0          | 25                  | 200                     |         |
| Trial 2 | 2.1          | 25                  | 200                     | 4.0     |
| Trial 3 | 2.2          | 25                  | 200                     | 3.9     |

5. Immediately add 1mL SOC-RM
  6. Transferred all medium and plasmids (liquid content) to the 15 mL tube
  7. Tape the lid while not completely fasten the tube as it is airtight
  8. Incubate at 37°C, 209 rpm for 3 hrs
  9. Put the agar plate into the incubator.
  10. Dilute the plasmid using LB medium with a ratio of 1: 1000 and then 1: 100 thus ending with a dilution of 1:1,000,000.  
→ for 1:1000 use 1  $\mu$ L: 999  $\mu$ L  
→ for 1:100 use 1  $\mu$ L: 99  $\mu$ L
  11. Use application stick to spread the diluted plasmid solution over the LB plates
  12. Take the rest to the 1:100 mixture and centrifuge it at 5000 g for 5 minutes  
→ remove the supernatant and add 100 microliters of LB medium to resuspend pellet & repeat step 11 again with new LB plate
- Done by Chloe W, Olivia Lin, Abigail, Abby T

**Date: 20260519**

**Title: Medium Preparation**

**Purpose: Prepare growth medium for *Bacillus subtilis***

**Procedure:**

1. Prepare the medium in 500 mL flask
    - LB broth: 6.25 g
    - D-sorbitol: 22.77 g → concentration to 0.5 M
    - Deionized water: Add 250 ml
  2. Stir on the heat plate until fully dissolved
  3. Autoclave for 20 minutes
- Done By Chloe W, Olivia Lin, Abigail, Abby T.

**Date: 20260519**

**Title: Electroporation competent cell (*Bacillus subtilis*) preparation**

**Purpose: To prepare the competent cell for electroporation**

**Procedure:**

1. Dilute the overnight culture of *Bacillus subtilis* to 16-fold in medium (GM medium).
2. Grow the *Bacillus subtilis* in 37°C, 200 rpm to OD600 of 0.915.
3. Cool the cells on ice for 10 min with shake.
4. Harvest the cells by centrifugal ion at 4°C and 5000 g for 5 min.
5. Wash the cells twice with EP buffer (add 1 ml and discard it).
6. Resuspend the cells in 1/40 volume of the culture volume with EP buffer.
7. Store the cells in -80°C.

Done by Gabriel, Recorded by Sophie L

2026/5/20 ✓

**Date: 20260520**

**Title: Miniprep**

**Purpose: Isolate and purify small amounts of plasmid DNA from a small culture of *Bacillus subtilis***

**Procedure:**

1. Pellet 1.8 mL bacterial overnight culture by centrifugation at >8000 rpm (6800 g) for 3 minutes at room temperature
2. Resuspend pelleted bacterial cells in 250  $\mu$ L Buffer P1
3. Add 250  $\mu$ L Buffer P2 and mix by inverting the tube 4-6 times until the solution becomes clear (DO NOT PIPETTE!!)
4. Add 350  $\mu$ L Buffer N3 & mix by inverting the tube 4-6 times
5. Centrifuge for 5 minutes at 23444 g
6. Apply 800  $\mu$ L supernatant from step 5 to column by pipetting. Centrifuge for 30 seconds & discard flow through. Repeat two times until the original solution is completely used up
7. Wash by adding 0.5 mL Buffer PB. centrifuge for 30 seconds and discard the flow through
8. Wash by adding 0.75 mL Buffer PE. centrifuge for 30 seconds and discard flow through
9. Centrifuge for 1 min to remove residual wash buffer
10. Place column in 1.5 mL microcentrifuge tube and add 50  $\mu$ L nuclease free water to the center of the column. Let it stand for 1 minute and then centrifuge it for 1 minute.

**Results:**

|          | Concentration (ng/ $\mu$ L) | A260/A280 | A260/A280 |
|----------|-----------------------------|-----------|-----------|
| Sample 1 | 216                         | 2.455     | 1.895     |
| Sample 2 | 291.9                       | 1.508     | 1.865     |

Done by Sophie Chen and Chloe. Recorded by Chloe

**Date: 20260520**

**Title: LB Agar plate**

**Purpose: Prepare agar plates for electroporation**

**Procedure:**

1. Formula as below:

| Components         | Mass   |
|--------------------|--------|
| Broth Powder       | 6.25 g |
| Agar powder        | 3.75 g |
| ddH <sub>2</sub> O | 250 ml |

2. Add one magnetic stirring bar to the flask
3. Place the flask containing the mixture on the hot plate
4. Remove the stirring bar from the flask
5. Cover the flask with aluminum foil and apply an indicator tape to ensure the mixture is successfully autoclaved
6. Place the flask in autoclave
7. Place the new petri dishes in BSC and turn on the UV light
8. Set up water bath at 55°C
9. Disinfect the working surface of the BSC and all equipment by spraying them with 70% ethanol.
10. Add 500 µL of antibiotics (ampicillin, chloramphenicol) into each corresponding flasks
11. Use serological pipette to add 10-15 ml of melting LB agar into the petri dish, avoid overfilling
12. Arrange the petri dishes in stacks inside the cabinet
13. Let the agar plates dry for at least 1 hour in the BSC

Done by: Abby T and Abigail, Recorded by: Abby T

2026/5/21 

**Date: 20260521**

**Title: Electroporation**

**Purpose: Testing the conditions by which has the best transformation efficiency on *Bacillus subtilis*.**

**Procedure:**

1. Prewarm the recovery medium at 37°C
2. Mix pNW33N-L2 (sfGFP) (216 µg/mL ,4.6 µL, 95.4 µL electrocompetent cells)
3. Pre-chill 0.2 cm cuvette with plasmid- competent cell mixture
4. Set up electroporator

|         | Voltage (kv) | Capacity ( µF) | Resistance (Ω) | TC (ms) |
|---------|--------------|----------------|----------------|---------|
| Trial 1 | 1.987/2.0    | 25             | 300            | 6.5     |
| Trial 2 | 2.092/2.1    | 25             | 300            | 6.7     |
| Trial 3 | 2.195/2.2    | 25             | 300            | 6.8     |

5. Immediately add 1 mL SOC-RM to the cuvette
  6. Transfer all medium and plasmid (liquid content) to the 15 ml tube
  7. Tape the lid while not completely fastening the tube as it is air tight
  8. Incubate at 37°C 209 pm for 3 hrs
- Done by Sophie L. Supervised by Chars

9. Put the Agar Plate into the incubator (for pre-warming)
  10. dilute the plasmid using LB medium with a ratio of 1:1000 and then 1:100 thus ending with a dilution of 1:1000,000
    - for 1:1000 (1 µL: 999 µL)
    - for 1:100 (1 µL: 99 µL)
  11. Use application stick to spread the diluted plasmid solution over the LB plates
  12. Take the rest of the 1:100 mixture and centrifuge at 5000 g for 5 minutes
    - remove the supernatant and add 100 µL of LB medium to resuspend pellet
    - repeat step 1 again with new LB plate
- Done by Ryan Yuan, Chloe W  
Supervised by Olivia Lin

2026/5/22



**Date: 20260522**

**Title: ½ MS Medium preparation**

**Purpose: Preparation for 5% growth medium**

**Procedure:**

1. Formula as below:

- ½ MS medium powder
  - Mass: 1.975 g
- Distilled water
  - Volume: 50 ml

2. Use a hot plate to stir until no sediments left in the solution

3. Wrap the neck with aluminium foil and stuck a piece of indicator tap on top

4. Autoclave for 20 minutes at 150°C

5. Pour the agar solution 1ml into the 1.5 ml tube lid with a pre-punched 2 mm hole.

6. Wait for the agar to solidify.

7. Place one Arabidopsis seed into the 2 mm hole of each of the 40 eppendorf tube caps.

Done by Abby K, Ethan Chang, Olivia Lin, Sophie H

2026/5/23



**Date: 20260523**

**Title: Electroporation**

**Purpose: Testing the conditions by which has the best transfer efficiency on *B. subtilis***

**Procedure:**

1. Prewarm the recovery medium at 37°C
2. Mix 4.63 µL of pNW33N-L2 (sfGFP) (concentration 216 µg/ml) and 95.37 µL of competent cell
3. Pre-chill 0.2 cm cuvette with plasmid-competent cell mixture
4. Set up electroporator with the following settings:

|         | Voltage (kV) | Capacity (µF) | Resistance (Ω) | TC (ms) |
|---------|--------------|---------------|----------------|---------|
| Trial 1 | 2.089/2.1    | 25            | 300            | 5.1     |
| Trial 2 | 2.280/2.3    | 25            | 300            | 4.4     |
| Trial 3 | 2.486/2.5    | 25            | 300            | 3.9     |

5. Immediately add 1ml of SOC-RM to the cuvette
  6. Transform all medium and plasmid to 15 ml tube
  7. Tape the lid while not completely fastening it airtight
  8. Incubate at 37°C at 200 rpm for 3 hours
- Done by Naomi, Sara, Phoebe, Ethan C, Sophie C, Sophia Y
9. Put the agar plate into the incubator to prewarm
  10. Dilute the plasmid using LB medium with a ratio of 1:1000 then 1:100 thus ending with a final dilution of 1:1,000,000
    - a. For 1:1000 use a ratio 1 µL : 999 µL
    - b. For 1:100 use a ratio of 1 µL to 99 µL
  11. Use application stick to spread the diluted plasmid solution over the LB plates
  12. Take the rest of the 1:100 mixture and centrifuge at 5000 g for 5 minutes
  13. Remove the supernatant and add 1000 µL of LB medium to resuspend the pellet
  14. Repeat step 11 again with LB plates containing chloramphenicol
- Done by Olivia Lin, Phoebe, Ethan Chang, recorded by Olivia Lin, Naomi Lin, Sarah Chou

**Problems encountered:**

- someone doing the first part of the experiment forgot to Label the voltage values onto the 15ml tube so we are not sure which is 2.3 kV and which is 2.5 kV
- The 2.1 kV solution (for some reason) had half the volume of the other two resulting in the same dilution factor but using 0.5 µL : 499.5 µL for 1:1000 ratio and 0.5 µL : 49.5 µL for 1:100 ratio

2026/5/24 

**Date: 20260524**

**Title: Agar Plate Preparation**

**Purpose: Preparation for agar plate**

**Procedure:**

1. Formula as below:

|                    | Amount   |
|--------------------|----------|
| ½ MS Medium Powder | 3.95 g   |
| Distilled Water    | 100 mL   |
| NaCl               | 0.5844 g |

2. Use a hot plate to stir (with a stirring rod) until no sediments remain in the solution.
3. Wrap the neck of the flask with aluminum foil and stick a piece of indicator tape on top.
4. Autoclave for 20 minutes at 127°C.
5. Wait for the pressure to reach 0 before removing the flask.
6. While waiting for the BSC to be ready, keep the flask in the water bath to prevent solidification.
7. Pipette 20 mL of agar solution into each plate. Repeat for a total of five plates.
8. Wait for the agar to solidify.
9. Defrost the trehalose.
10. Disinfect the spatula by dipping it into fuel alcohol then heating it with a Bunsen burner.
11. Wait for the spatula to cool.
12. Pipette the respective volumes of trehalose for different molarities. Do one plate for each.

| Molarity (Trehalose Concentration) | Volume (Stock: 100 mM) |
|------------------------------------|------------------------|
| 0.125 mM                           | 25 µL                  |
| 0.25 mM                            | 50 µL                  |
| 0.5 mM                             | 100 µL                 |

13. Spread the trehalose with the spatula until the spreading is not smooth.
14. Disinfect the spatula and repeat for each molarity.

Done by Chloe W, Ethan C, and Sophia Y

2026/5/25 

**Date: 20260525**

**Title: Agar Plate Preparation**

**Purpose: Preparation for agar plate**

**Procedure:**

1. Formula as below:

|                    |          |
|--------------------|----------|
| ½ MS Medium Powder | 5.925 g  |
| Distilled Water    | 150 mL   |
| NaCl               | 0.8766 g |

Add the contents into a flask.

2. Use a hot plate to stir (with a stirring rod) until no sediments are left in the solution.
3. Wrap the neck with aluminum foil and stick a piece of indicator tape on top.
4. Autoclave for 20 minutes at 127°C.
5. Wait for the pressure to reach 0 before retrieving the flask.
6. While waiting for the BSC to be ready, keep the flask in the water bath to prevent solidification.
7. Transfer 20 mL of agar solution into each plate. Repeat for a total of five plates.
8. Wait for the agar to solidify.
9. Thaw the trehalose.
10. Disinfect the spatula by dipping it into fuel alcohol then heating it with a Bunsen burner.
11. Wait for the spatula to cool.

Done by: Chloe, Olivia L, Sophie L, Sara C, Recorded by Sophie L

**Date: 20260525**

**Title: Competent Cell**

**Purpose: Making Competent Cell for Electroporation**

**Procedure:**

1. Dilute an overnight culture of *Bacillus subtilis* in 16 fold in growth medium (LB medium containing 0.5 M growth medium) (313 µL B. sub. In 5 ml growth medium), grow at 37°C to an OD600 of 0.851
2. Cool the cells on ice and put the ice bucket to harvest cells
3. Centrifuge at 4 degree at 5000 g for 5 min to harvest cells
4. Add 1 µL of ice cold electroporation medium to resuspend cell in 1.5 ml microtube, then centrifuge at 5000 g for 1 minute and discard supernatant
5. Repeat 4 times to wash the cells
6. Suspend the cells in 1/40 of the culture volume of the electroporation solution (0.125 ml)
7. Store competent cell at -80°C

Done by Chloe, Olivia Lin, Sophie Liu, Sara Chen Supervised by Gabriel

2026/5/26 

**Date: 20260526**

**Title: Electroporation**

**Purpose: To test/optimal voltage for enhancing the electroporation efficiency**

**Procedure:**

1. Mix the plasmid pNW33N-L2 (sfGFP) and the competent cell (5/25) Plasmid (13µL) + competent cell (287 µL) (216 ug/µL).
2. Transfer 100 µL to each cuvette (0.2 cm).
3. Ice bath the cuvette for 3 min.
4. Set up and start the electroporation process.

|           | Voltage (kV) | Capacity (µF) | Resistance (Ω) | TC (ms) |
|-----------|--------------|---------------|----------------|---------|
| Cuvette 1 | 2092         | 25            | 300            | 7.3     |
| Cuvette 2 | 2285         | 25            | 300            | 7.3     |
| Cuvette 3 | 2493         | 25            | 300            | 7.2     |

5. Immediately add 1 mL warm recovery medium to the cuvette and transfer to the tubes.
6. Incubate at 37°C, 200 rpm for 3 hr.
7. Prepared 3 LB plate and 3 LB plate with Chlorophenicol.
8. Prepare the dilute samples of each in 1:1000000.
9. The remaining samples were then centrifuged in 5000 g for 5 min.
10. The dilute samples were spread in the LB plate.
11. The remaining samples were spread in the LB plate with Chlor.
12. Incubate in 37°C overnight.

Done by Gabriel, Recorded by Sophie L

**Date: 20260526**

**Title: Seeding arabidopsis in the agar plate**

**Purpose: Sow seeds on agar plate**

**Procedure:**

1. Volume

| Molarity (Trehalose Concentration) | Volume (Stock: 100 mM) |
|------------------------------------|------------------------|
| 0.125 mM                           | 37.5 µL                |
| 0.25 mM                            | 75 µL                  |
| 0.5 mM                             | 150 µL                 |

2. Spread the difference volume of ½ Ms medium on the ½ M agar plate
3. Seeding 9 arabidopsis seeds on each agar plate, separate with equal interval

4. using aluminum foil cover it and label the seeding time and the time it nee to taken out ( 3 days after)
5. Putting into 4 °C and wait for 3 days vernalization

Done by Sarah Chou

**Date: 20260526**

**Title: Hydroponic medium preparation**

**Purpose: Prepare hydroponic medium**

**Procedure:**

1. Measure 2.3 g  $\frac{1}{2}$  MS medium in 1000 mL de-ionized water( mix with stirring bar)
2. Determine pH of the solution using a pH meter  
Before pH:4.0 / after pH :5.67
3. Divide 1000 mL solution into two 500ml of bakersAutoclave for 20 min at 127°C
4. Fill the hydroponic box with medium and transplant the seedlings into the box.

Done by Chloe W , Record by Sarah Chou

2026/5/27



**Date: 20260527**

**Title: Hydroponic Medium Preparation**

**Purpose: Prepare medium for Hydroponic system**

**Procedure:**

1. Measure 1.15 g  $\frac{1}{2}$  MS medium powder into 1000 ml flask with 500 ml deionized water.\*3 (medium#1, medium#2, medium#3)
2. Mix the medium with stirring bar
3. Determine the pH of the solution via using a pH probe
4. Measure the pH and adjust it with KCl

| Medium | Before adjust (pH) | After adjust (pH) |
|--------|--------------------|-------------------|
| #1     | 4.09               | 5.02              |
| #2     | 4.06               | 5.64              |
| #3     | 4.08               | 5.56              |

5. Autoclave pH-adjusted medium at 121°C for 20 mins.
  6. Let the three medium cool down in room temperature
- Done by Sophie L, recorded by Sophie H.

**Date: 20260527**

**Title: Miniprep**

**Purpose: experiment abort because unable to resuspense pellet properly**

**Procedure:**

1. Transfer 1.8 mL TB medium to 2 ml microtube repeat 7 time
2. Centrifuge at 16000g for 1 minute \*balance microtube
3. Discard the supernatant
4. repeat step 1 until collected all medium
5. Add 200  $\mu$ L MX1 buffer (store at 4°C)
6. Resuspend the pellet by pipetting

\*Experiment abort because unable to resuspend pellet properly

Done by : Sophie C , Sophie L , Sophia Y , Alex  
Supervised by Venus

2026/5/28





2026/5/29 

**Date: 20260529**

**Title: Preparation LB agar plate**

**Purpose: Preparation LB agar plate**

**Procedure:**

1. Formula as below:

|                 | mass(g) |                      | mass(g) |
|-----------------|---------|----------------------|---------|
| Broth powder    | 10      | Broth powder         | 10      |
| Agar powder     | 6       | Agar powder          | 6       |
| Distilled water | 400 mL  | Distilled water      | 400     |
|                 |         | Kanamycin (50 mg/mL) | 0.4 ml  |

2. Add broth , agar powder, distilled water in a flask
3. Stir until no sediment
4. Wrap the neck with aluminum foil and stick a piece of indicator tape on top to ensure the mixture is successfully autoclaved
5. Place the flask in autoclave
6. Set up water bath at 55°C
7. While waiting for the BSC to be ready , keep the flask in the water bath to prevent solidification
8. Add 400 µL kanamycin
9. Using pipette add to aliquot 20 mL of agar solution into each plate , repeat for all plate
10. Wait for agar to solidify

Done by Sarah C , Sophie H , Olivia L, Ryan Y , Sophia Y

2026/5/30



**Date: 20260530**

**Title: Agar Plate Preparation**

**Purpose: Preparation of agar plates for *Arabidopsis* planting**

**Procedure:**

1. Formula as below:

|                    |           |
|--------------------|-----------|
| ½ MS Medium Powder | 0.5925 g  |
| Distilled Water    | 15 mL     |
| NaCl               | 0.13149 g |

2. Use a hot plate to stir (with a stirring rod) until no sediments are left in the solution.
3. Wrap the neck with aluminum foil and stick a piece of indicator tape on top.
4. Autoclave for 20 minutes at 127°C.
5. Wait for the pressure to reach 0 before retrieving the flask.
6. While waiting for BSC, keep the flask in the water bath to prevent solidification.
7. Pipette 500 µL agar solution into each well of 24-well plates. Repeat for 2 well plates.
8. Wait for agar to solidify.
9. Repeat previous steps to prepare another spare.
10. Thaw the trehalose.
11. Add five 2.5 mm glass beads into each well from rows B–D (inside the BSC).
12. Pipette respective volumes of trehalose for different molarities (50 µL per well).

|       | Molarity (Trehalose Concentration) |
|-------|------------------------------------|
| Row B | 0.25                               |
| Row C | 0.50                               |
| Row D | 0.75                               |

13. Shake the well plates until trehalose dries.

Done by: Sophie L, Chloe, Sara Chen, Recorded by Sophie L

**Date: 20260530**

**Title: LB Medium Preparation**

**Purpose: Prepare Bioreactor Medium to culture *Bacillus Subtilis***

**Procedure:**

1. Formula as below:

|                    |        |
|--------------------|--------|
| LB broth powder    | 12.5 g |
| ddH <sub>2</sub> O | 500 mL |

2. Add one magnetic stirring rod to the flask.
3. Place the flask containing the mixture on the hot plate, and stir until dissolved.
4. Remove the stirring rod from the flask.
5. Cover the flask with aluminum foil and apply an indicator tape to ensure the mixture is successfully autoclaved.
6. Place the flask in autoclave.

Done by: Sophie L, Chloe, Recorded by Sophie L

**Date: 20260530**

**Title: Agar Preparation**

**Purpose: preparation of agar for hydroponics**

**Procedure :**

1. Formula as below:

|                    |       |
|--------------------|-------|
| ddH <sub>2</sub> O | 10 mL |
| ½ MS Medium Powder | 0.4 g |

2. Add the contents into a beaker.
3. Use hot plate to stir (with a stirring rod) until no sediments are left in the solution.
4. Wrap the neck with aluminium foil and stick a piece of indicator tape on the top.
5. Autoclave for 30 min at 127°C.
6. Wait for the pressure to reach 0°C before retrieving the flask.
7. Pipette 800 µL of agar solution into each of the ten tips in the laminar flow.
8. Wait until the agar solution solidifies and place Arabidopsis seeds on all of the tips.
9. Cover them with aluminium foil and place it in the fridge for 2-10 days.

Done by Ryan Wei, Alex

2026/5/31 

**Date: 20260531**

**Title: Agar Plate Preparation**

**Purpose: To observe whether the pH of agar has an effect on *Arabidopsis* growth**

**Procedure:**

1. Formula as stated:

|                    |        |
|--------------------|--------|
| ddH <sub>2</sub> O | 200 mL |
| ½ MS Medium Powder | 7.9 g  |

2. Mix the contents in a conical flask.
3. Use a stirring bar to mix evenly until no sediments are left in the flask (on a hot plate).
4. Wrap the neck with aluminum foil & stick a piece of indicator tape on top.
5. Autoclave for 30 min at 127°C.
6. Wait for the pressure to reach 0 before retrieving the flask.
7. Dispense 30 mL of agar solution into each of the 6 plates in the laminar flow.

Done by Chloe, Professor Kyle, Recorded by Sophie L